Project: Testing Logit-Based Metrics of LLM on Out-of-Domain Knowledges: Impact of Gradient Dynamics and RAG/ICL Methods

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I. Preface

This project started before the proposal, but for the final report, I still have to change the specific goal. Here is why:

The original goal was to use the Variational Autoencoder (VAE) to map the token-level or sentence-level epstemic uncertainty (EU) score[1] into latent space. Recall the rationals:

- The uncertainty estimation itself is uncertain (and should be in distribution form)
- Utilize KL to pull the estimations to the prior so that we can have a reference, as well as to smooth the latent estimation space
- The latent space's distribution form innately constrains the upper and lower bounds of the estimations

It seems viable, but in experiments it consistently failed to converge until I had to stop because of the limited time. I suspect that this stems from the overly simple model architecture. However, attempting to deepen or widen the VAE could lead to posterior collapse and will certainly break the near-linear interpretable property of the decoder, thus compromising the intended explainability of the latent space.

The old goal of uncertainty estimation can still be achieved in a point-estimation way. With a setup inspired by the "IDK tuning" work[2] (on Pythia model 2.8b over The Pile's 1b subset), I plotted the EU curve and observed a pronounced single peak. From analysing either the peak EU position or the shaded area under the EU curve, we can fit a bound-scaler leading to a usable metric.

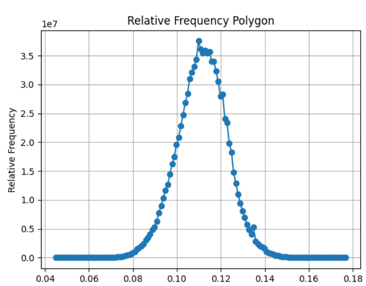


Fig. 1: EU curve on Pythia over The Pile

I also found that a representative model, LLaMA 3.1, already shows an exceptionally strong answer-refusal ability, a repeatedly shown pattern of outputing "I do not…", "I am unable to…" against out-of-domain questions. This is unlikely to arise from generic web-crawled pretraining data and instead I suspect that it may be introduced from the supervised fine-tuning (SFT) stage. Even after I manipulated the system prompt to spoof its knowledge cutoff date, the model persisted in a similar refusal phrasing, perhaps hinting that hidden-states implicitly encode metrics like EU in a manner similar to Implicit CoT[3]. However, this phenomenon lies beyond the scope of my current work.

In short, for the reasons mentioned above, this project ultimately chooses to empirically verify the properties of EU/logit-as-evidence-strength metrics rather than designing a new EU-based answer-refusal method.

II. Introduction

Although epistemic uncertainty (EU)[1] is not an objective directly optimised during model training, the original EU paper proposed EU as a measure of uncertainty driven by knowledge gaps, explained by gradient dynamics. However, that paper offered mostly theoretical analysis, no experiments were conducted to verify this gradient dynamics phenomenon, and few are conducted to establish the reliability of EU/logit-based score in practice. Moreover, contemporary retrieval-augmented generation (RAG) methods achieve substantial accuracy gains without invoking any gradient dynamics. And there exists paper that argues the similiarity between RAG/In-context-learning and gradient-based training[4]. Thus, how would EU/logit-based score work in non-gradient-dynamics scenario is also interesting.

In summary, the key research questions are:

- Is EU/logit-based metrics directly related to the accuracy of the model output? (Reliability)
- Does gradient dynamics truly affect EU/logit-based metrics in the proposed way? (Explainability)
- How will logits change when applying methods without gradient dynamics, like RAG? Does it still affect metrics like EU in a similar way to gradient dynamics? (Generalization)

Additionally, if EU/logit-based score truly provides a robust uncertainty score tied to knowledge coverage (reflected in the above properties), it may serve as a powerful lens for studying LLM's learning behaviour. For example: whether supervised fine-tuning (SFT) can inject knowledge that generalizes beyond its specific training instances: Unlike reinforcementlearning–based methods, SFT allows arbitrary, human-crafted target examples that can force the model to memorize new facts. If there're only changes in the question format, does SFT's forced knowledge injection still work?

Besides answering above questions, the contribution of this project also includes a novel dataset: To facilitate evaluation of out-of-domain question answering, this project introduces a carefully curated benchmark/dataset designed specifically to test LLM's out-of-domain knowledge coverage.

III. Dataset

The dataset part is the most important thing in this project setting. We need to make sure the testing knowledge is out-of-domain, simulating the most extreme testing scenario for uncertainty metrics, and making valid comparison before and after applying LLM learning methods. Existing datasets like Wiki-QA[5] are old and highly likely to be included in modern LLM's training dataset. The goal of this part is to find a way to build a dataset with affirmatory out-of-domain knowledges for recent LLMs.

To achieve this, we need to find a target model group first. There's an implicit balance between knowledge cutoff date and model's preformance. Too new models are harder to find enough out-of-domain knowledges for, while too old models are just bad and may sabortage the experiment's results due to bad instruction following. A good balanced separating date may be the end of 2023, when LLaMA 3.1 came out. It is widely used, representative, and most importantly, not distilled, so logit-based scores can stay accurate. It demonstrates the typical technologies of that period.

With a cutoff date of the end of 2023, I managed to find a dataset crawled from Wikipedia, with meta-information provided. Its cutoff date is September, 2024. With the "date_created" key, I extracted over 20000 items of abstract, item name, etc., filtered to be only created after 1st, January, 2024. They are likely 2024's newly happened things, and the text patterns are definitely not included in 2023's models' (like LLaMA 3.1's) training corpus.

Then I create the actual dataset in these steps:

1) Provide a system prompt of

```
You are given a short Wikipedia abstract
    delimited by <ref></ref>.
1. Devise ONE question that someone could
    answer **only if** they've read that
    abstract.
2. Provide the **correct answer** in one
    sentence.
Return JSON strictly in this form:
{
  "question": "...",
  "answer":   "..."
}
```

to ChatGPT-4o, generating about 20000 rough outputs. The references are the wiki abstracts.

2) Sometimes LLMs won't follow the instruction and output nonsense or out of format contents. So I removed any content around the first pair of {} and read and verify it with json. Invalid data are discarded.

3) Besides making sure the json has the needed keys, the generated values to these keys can be problematic. For example, the input abstract is way too simple, so only repeat-the-abstract kind of questions can be generated, and Wikipedia's abstract sometimes leads to redirect pages, which exist in the crawled data. So I checked the keywords like "abstract" or "redirect" in the values, and filtered these invalid questions out.

4) These questions are text-based, not easy to be automatically judged with simple rules and make logit selection extra hard. So I use this prompt

```
You are an assessment-item writer.

You will receive a single-object JSON
    containing:
{
  "question": "<original question text>",
  "answer":   "<the single correct
      answer>"
}

Augment the answer into a four-option
    single-correct item and return
    **only** a new JSON object with
    **exactly** these keys:

{
  "options":  ["A. <choice 1>", "B.
      <choice 2>", "C. <choice 3>", "D.
      <choice 4>"],
  "correct_choice": <the sole letter of
      the correct choice in the array
      (A/B/C/D)>
}

Rules for constructing the 'options' array
1. Include the given correct answer once;
    generate **three unique, plausible
    distractors** that:
   match the type/length/tone of the
       correct answer
   are factually incorrect for the
       question
   are not giveaways like "None of the
       above"
2. Randomly shuffle the four choices so
    the correct answer's position varies.
3. Keep each option concise (< 30
    characters).
4. Do **not** repeat the correct answer,
    and do not output explanations.

Output must be valid JSON (no markdown
    fences, headings, or extra keys).
Return nothing except that JSON object.
```

to generate multichoices with ChatGPT-4o.

5) Check format and clean up, now we have a multichoices dataset with 10387 items.

6) Manually input about 1% of them randomly into ChatGPT-o4-mini with Internet-search enabled and ex-

aminate. The dataset has a 100% pass rate.

7) Randomize the option letters of the questions and answers.

Here is a sample from the final dataset:

```
{
  "index": "336",
  "name": "2024 Utah State Treasurer
      election",
  "ref": "<ref>The 2024 Utah State Treasurer
      election is scheduled to take place on
      November 5, 2024, to elect the next
      state treasurer of Utah. Incumbent
      Republican treasurer Marlo Oaks won a
      special election after David Damschen
      resigned. Oaks is running for a full
      term.</ref>",
  "input": "Who is the incumbent Republican
      treasurer running for a full term in
      the 2024 Utah State Treasurer
      election?",
  "options": ["A. Marlo Oaks", "B. John
      Smith", "C. Emily Davis", "D. Sarah
      Johnson"],
  "output": "A",
  "answer": "Marlo Oaks is the incumbent
      Republican treasurer running for a full
      term."
}
```

IV. SETUP

Due to reasons mentioned in the last section, I chose LLaMA 3.1-8b-instruct as the model to use. For comparative experiments, there're 3 models in total, the first one is original LLaMA 3.1, the second is LLaMA 3.1 SFT-trained on the whole dataset for multichoices task, the last one is trained on the whole dataset for text answer generation task. All of the training sessions use the same hyperparameters of 1 epoch, rank 8 lora, cosine lr-scheduler with 0.1 warmup ratio, 3e-4 learning rate with Adam optimizer, batchsize 8*8GPUs=64. For multichoices task, there's a special system prompt to guide the model to output one letter out of A/B/C/D only. No reference was provided in training, for the purpose of injecting knowledge instead of encouraging copying.

I ran prediction with these models using greedy decoding on the proposed dataset and obtained these metrics:

- prediction, which is the output text
- eu, which is EU score for each token with the recommanded k=64 setting[1]
- choices logit, which are each token's A/B/C/D letter logits
- choices prob, which are each token's A/B/C/D letter probs
- the output id that has top prob, its logit and its prob
- mean/std/median/25% quantile/75%quantile of all logits from each token

Since we want to measure the model's confidence of its choice, the measured choice token has to be generated naturally. Only in this way can we get the real logit metric that can reflect the model's knowledge instead of the confidence of some intermedate tokens. Therefore, I used greedy decoding. Sometimes the model doesn't output A/B/C/D as the first token. In fact, in these cases, the logit evidence of A/B/C/D in that position is relatively, abnormally low. So we have to find the first A/B/C/D's id in the whole sequence, get its index, and obtain corresponding eu, choices logit, choices prob, top token, top logit top prob, and mean/std/median/25% quantile/75%quantile of all logits from that position.

Since the behaviour of the tokenizing, though it's rare, sometimes the chosen A/B/C/D from the sequence might be just part of some words. So I cross-compare this id (its corresponding letter) with the one cleaned from the output text to fairly judge the model's outputs. If no id of A/B/C/D is found, the first token will be used by force. This is tricky with the model trained with text output task, because its instruction following and output format is a little broken, although I have max token num to generate limited.

```
Examples:

 "prediction": "D."
 "prediction": "C. A pigment used in Roman
    mur"
 "prediction": "Gunnar Teodor Krantz served"
 "prediction": "A."
 "prediction": "D. Ideas linked to wokeness"
```

Luckily, most first tokens' A/B/C/D logits are high, hinting the model is still choosing from A/B/C/D.

After judging, the final metrics we get are (I will use these shorten terms in the resting pages):

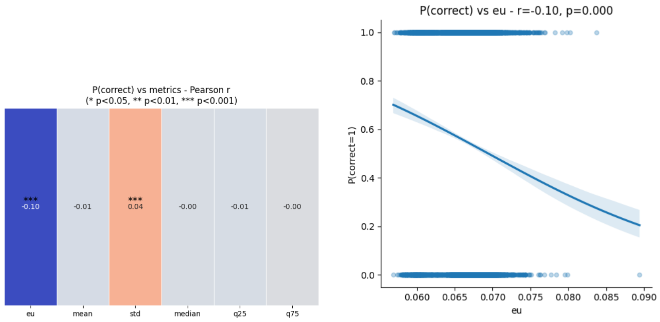
- correct, true or false (1 or 0)
- eu, which is EU score for the chosen token with the recommanded k=64 setting[1]
- logit, the ground truth letter's logit from the chosen token
- prob, the ground truth letter's prob from the chosen token
- logit_top, top logit from the chosen token
- prob_top, top prob from the chosen token
- mean/std/median/q25/q75 of all logits from the chosen token

V. EXPERIMENT AND EVALUATION

A. *Reliability*

First, we need to know whether EU/Logit-based score is reliable - when encountering out-of-domain questions, before and after applying learning methods. The original paper[1]'s experiments are lacking in these tests. I compute the Pearson r of correct versus logit-based metrics including eu/mean/std/-median/q25/q75 and display the results here. The latter metrics are mean to represent the logit distribution.

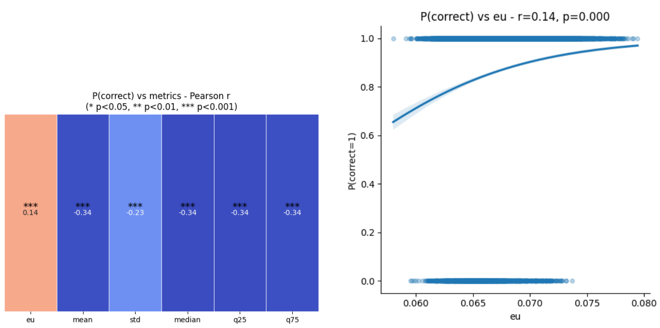
Test 3 cases: Not trained model without reference (knowledge absense scenario), multichoices task trained model without reference (gradient dynamics scenario), not trained model with reference (RAG/ICL scenario).



(a) heatmap (b) data points scatter & curve

Fig. 2: NoReference NoTrain

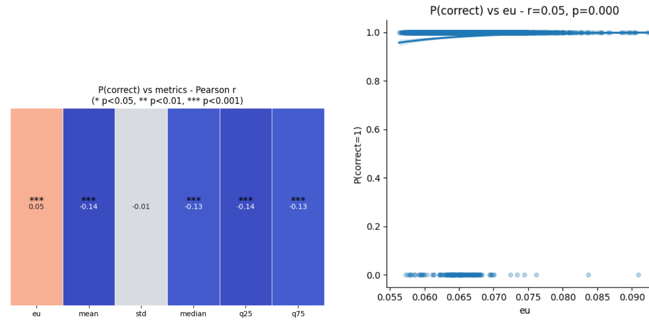
When not providing reference, it's obvious that EU score is negatively correlated with correct. It confirms that in the knowledge absence scenario, EU score is a good metric for uncertainty measurement/answer refusal.



(a) heatmap (b) data points scatter & curve

Fig. 3: NoReference Train

With the model trained on the exact same data as testing, strange things happened. EU is supposed to be the measurement of uncertainty, but it is now positively correlated with correct and all other logit metrics are negatively correlated with correct. Both are significant. Note that SFT is not the distillation operation mentioned in the original EU paper - the latter means learning the logit distribution by force. So there's no exception. SFT still uses softmax to alter logits, thus confirms the gradient dynamics description in that paper. The only difference between it and the pretraining is that SFT is given n fixed prompt token, which theoretically should not be the reason to lead the result to the opposition. So, this is truly weird.



(a) heatmap (b) data points scatter & curve

Fig. 4: Reference NoTrain

When providing the reference without training, similar thing happened. EU is positively correlated to correct, but this time it's less significant.

There's one possible explanation to this. In the last two cases, mean/median/q25/q75 all show that other logits are negatively correlated with correct, which indicates there is possibility only the correct word's logit or very few candidates' logits are raised, while the majority of words' logits are lowered. When selecting too large k, like k=64 as recommanded, the harder the learning method lowers the incorrect words' logits, the higher EU becomes. If so, these results didn't break the theory that the EU paper proposes. Still, logits-based scores like EU are proven not always reliable, at least depending on the right hyperparameters like k.

B. *Explainability and Generalization*

With the strange things above, the explainability of EU/logits-based metrics is even more important. We will now use NoReference_NoTrain as baseline, compare how the averaged metrics will change before and after providing reference, training, or both, and try to explain.

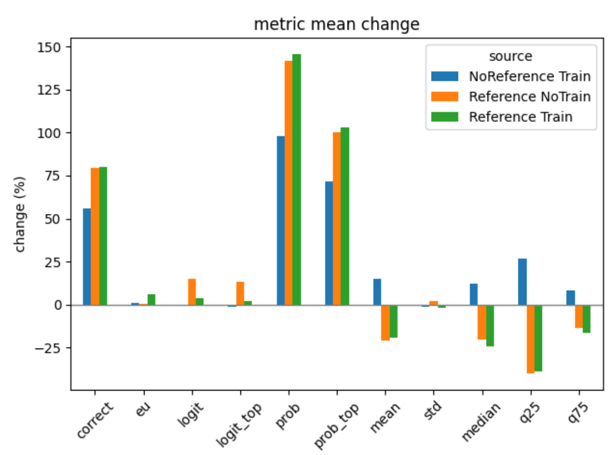


Fig. 5: Metric mean changes with NoReference_NoTrain as baseline

We can see that providing reference to trained model works the best, and providing reference (RAG) actually works better than 1 epoch training on the same dataset. The RAG method's results actually confirm the assumption made in the last subsection. It raises correct word's logit and lowers incorrect words' logits, just as the EU paper says what the cross-entropy gradients will do, which is very interesting. SFT trained model's metrics are extremely unexpected though. It seems that training will increase the values of lots of logits, if not all, which is against the gradient dynamics explaination from the EU paper[1] ("for misclassified samples, the $\log \sum_j e^{z_j}$ term exerts downward pressure on *all* logits, whereas the $-y_j z_j$ term selectively boosts the true-class logit"). Last, with reference provided, the trained model's logit distribution starts to behave more as expected.

In the SFT case, top logit/predicted logit barely increases, while overall logits are boosted. The change of the logit vector direction (to the right direction) cannot be larger than RAG/ICL's. Logit-based metrics can show us this.

In summary, the explaination to the EU score made by the original paper is doubtful, at least not universal (at least not suitable for similar short SFT setup). And EU may not be an absolutely correct measure of uncertainty, however, logit-based metrics on their surface can still show us the effectiveness of non-gradient-dynamics learning methods like RAG/ICL, which is very useful.

C. Results for the SFT knowledge injection example

Though the shocking results we got above, and EU is not fully trusted, I'll still show the results of the example in Introduction section (training on text output task, testing on multichoices task, with same knowledges, to research SFT's question format generalization ability). To be clear, the results here are not as trustable as the above two subsections, because the model's outputs are a lot more likely to be in text format, without giving an option letter, especially when there's no reference provided, so being forced to use the first token as the choice token will happen frequently.

We can see, the logits are boosted even more, and providing reference is not able to reverse this trend. Correct/ground truth's logit/prob/etc. are lower. Probably due to the model's instruction following is broken. The training on text output task did nothing good here. If using the same logit vector direction comparison approach to review this test, then this is the worst scenario (exactly the opposite effects to RAG).

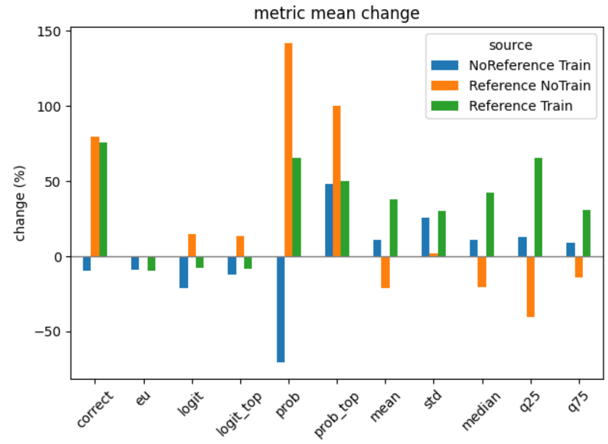


Fig. 6: Metric mean changes with NoReference_NoTrain as baseline, using TextTrain for SFT

VI. Conclusion

After these experiments, we have really unexpected results. Non-gradient-dynamics methods work just like the gradient theory shows while gradient-dynamics method does not. I acknowledge that this project is lacking of further experiments (pretraining, different hyperparameters, different models, multiple question types (such as using LLM as a judge for text test), etc.), due to time constraints, extremely poor server connectivity, and other unavoidable reasons. But we should still be cautious to non-optimization-target logits-based uncertainty scores like EU, at least doubt them when doing SFT training (with few epochs and small datasets).

It also pointes us several future directions: How does RAG/ICL alter logits inside models? Is it really because dot-production of self-attention[4]? And of course, why does SFT behave like this? Will that last if hyperparameters are changed? Will that happen with pretraining?

I will dive deeper into these research questions. And hopefully I'll get some interpretable results. Plus, the Wiki dataset built for this project will be open-sourced.

References

[1] Huan Ma et al. *Estimating LLM Uncertainty with Evidence*. 2025. arXiv: 2502.00290 [cs.CL]. URL: https://arxiv.org/abs/2502.00290.

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